Regression and Residuals

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

One readable numerical-error record.

Trace residuals,compare signed and absolute means and preserve missing or extreme evidence.

Keep one habit: W1–W3/W5 → selected fresh pairs → numerical side-project record → changed-case return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Error Analysis. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Accuracy and Imbalance. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Train a Classifier. Give an example and a mistake that the idea helps you avoid.

Error Analysis

> **My remembered idea, example and correction**

Accuracy and Imbalance

> **My remembered idea, example and correction**

Train a Classifier

> **My remembered idea, example and correction**

W1 · Explain the mechanism

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main matched observed heights4,7,9cm/predictions5,6,9:residuals-1,+1,0cm,mean0,MAE2/3cm. Picture observed5,8,12/predicted7,7,10:residuals-2,+1,+2,mean1/3,MAE5/3;raising predictions1cm gives-3,0,+1,mean-2/3,MAE4/3. Current lab recordsR1(3,4),R2(5,4),R3(6,8),R4(10,8),R5(8,12),R6(16,12) observed/predictedcm:residuals-1,+1,-2,+2,-4,+4,mean0,MAE7/3.

  Negative means overprediction;positive underprediction;zero exact on the pair.

- Reference for this exercise: open this week's student lesson, "Regression and Residuals", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- Compute the worked **residual**s and identify **overprediction**.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Trace the current experiment

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main matched observed heights4,7,9cm/predictions5,6,9:residuals-1,+1,0cm,mean0,MAE2/3cm. Picture observed5,8,12/predicted7,7,10:residuals-2,+1,+2,mean1/3,MAE5/3;raising predictions1cm gives-3,0,+1,mean-2/3,MAE4/3. Current lab recordsR1(3,4),R2(5,4),R3(6,8),R4(10,8),R5(8,12),R6(16,12) observed/predictedcm:residuals-1,+1,-2,+2,-4,+4,mean0,MAE7/3.

  Negative means overprediction;positive underprediction;zero exact on the pair.

- Why does mean **residual** zero not imply perfect **prediction**s?

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Repair the reasoning

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main matched observed heights4,7,9cm/predictions5,6,9:residuals-1,+1,0cm,mean0,MAE2/3cm. Picture observed5,8,12/predicted7,7,10:residuals-2,+1,+2,mean1/3,MAE5/3;raising predictions1cm gives-3,0,+1,mean-2/3,MAE4/3. Current lab recordsR1(3,4),R2(5,4),R3(6,8),R4(10,8),R5(8,12),R6(16,12) observed/predictedcm:residuals-1,+1,-2,+2,-4,+4,mean0,MAE7/3.

  Negative means overprediction;positive underprediction;zero exact on the pair.

- Interpret **residual**s -2, 0, 3 ordered by growth stage.

> **My reasoning and evidence**
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MAIN LESSON · SECTION 16

W5 · Plan the next check

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main matched observed heights4,7,9cm/predictions5,6,9:residuals-1,+1,0cm,mean0,MAE2/3cm. Picture observed5,8,12/predicted7,7,10:residuals-2,+1,+2,mean1/3,MAE5/3;raising predictions1cm gives-3,0,+1,mean-2/3,MAE4/3. Current lab recordsR1(3,4),R2(5,4),R3(6,8),R4(10,8),R5(8,12),R6(16,12) observed/predictedcm:residuals-1,+1,-2,+2,-4,+4,mean0,MAE7/3.

  Negative means overprediction;positive underprediction;zero exact on the pair.

- A report combines 2 cm and .02 m as if they were different magnitudes. Repair it.

My reasoning and evidence

MAIN LESSON · SECTION 4

Try alone · Selected experiment task

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Pair the new three observed and predicted heights in their supplied order, keeping cm throughout. Use observed minus predicted, then explain each sign before averaging. The independent pairs differ from the pictured values even if a resulting residual pattern happens to match; show subtraction evidence rather than copying a pattern. No fitting data or future-performance result is supplied.

- Observed [6, 10, 13], predicted [8, 9, 11] cm. Give **residual**s, mean **residual** and error directions.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 9

Project · Keep the experiment evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Use a miniature growth-**prediction** side experiment to distinguish **regression evidence** from the main classifier metrics.
- The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

MAIN LESSON · SECTION 10

Finish · Judge your evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- - I can explain the mechanism.
  - I can show evidence.
  - I can handle a changed case.

- I can name a limit.

  Mark each as not yet, with help or independently, and attach supporting work.

- Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Use W2: Explain cancellation for **Regression** and **residual**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- Use the worked case for **Regression** and **residual**s. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier learning target

REFERENCE · SECTION 5

Reference · Earlier learning target

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- Actual leaf length is 7 cm and **prediction** is 10 cm. Using **residual**=actual−predicted, calculate **residual** and absolute error. Interpret the sign.

▼ **Extra practice · Earlier complete case**

REFERENCE · SECTION 6

Extra practice · Earlier complete case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Actual height=12 cm, **prediction**=9 cm. **Residual** means actual minus predicted.
- Calculate and interpret.
- Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ Extra practice · Changed case

REFERENCE · SECTION 7

Extra practice · Changed case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Define **residual** as actual minus predicted. Actual height is 14 cm and **prediction** is 17 cm. Calculate **residual** and interpret its sign.

- My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 19 Test and reflect

▼ Extra practice · Design a revealing test

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Extra practice · Earlier transfer**

REFERENCE · SECTION 11

Extra practice · Earlier transfer

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Actual leaf length is 7 cm and **prediction** is 10 cm. Using **residual**=actual–predicted, calculate **residual** and absolute error. Interpret the sign.

- Record support used: none, hint or teacher explanation.

My independently reasoned answer

▼ **Reference · Earlier W4 record**

REFERENCE · SECTION 15

Reference · Earlier W4 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Observed [6, 10, 13], predicted [8, 9, 11] cm. Give **residual**s, mean **residual** and error directions.

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 18

Extra practice · Pictured case

> Keep this task's complete inputs, stated rule and support record together.

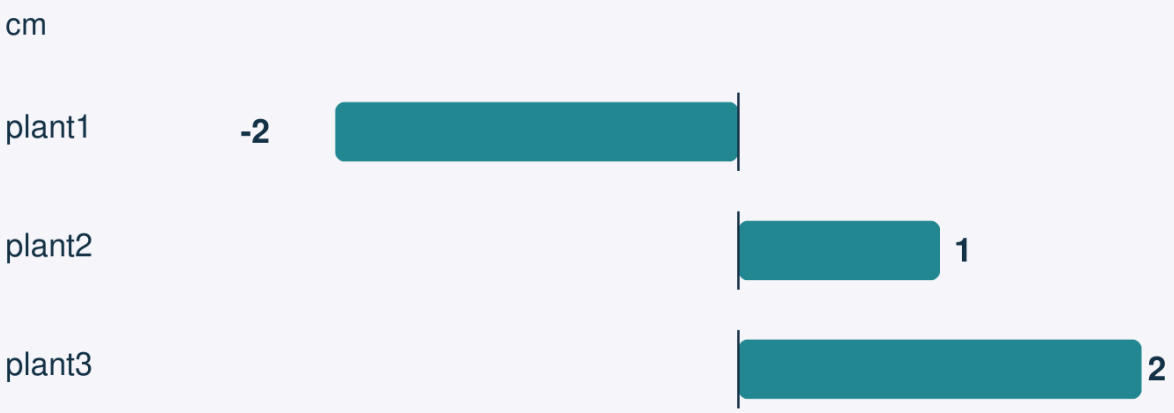
Task and evidence record

- Use **residual** signs to say whether each plant-height estimate is too high or low.
- **Residual**=observed-predicted. Observed heights [5, 8, 12] cm; predicted [7, 7, 10] cm.
- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.
- Which statistic reveals the size of errors without cancellation? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.
- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

AI ACADEMY / AI-4/19

Observed minus predicted height



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-4/19

Regression and Residuals

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Subtract matched pairs

Residuals are5-7=-2, 8-7=1, 12-10=2 cm.

2. Read directions

First prediction is too high by2; second too low by1; third too low by2.

3. Avoid cancellation

Mean residual is 1/3 cm; MAE is 5/3 cm, so small signed mean does not imply small errors.

CHANGE ONE THING

Raise all predictions by1 cm.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 19

Reference · Project portfolio milestone

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Project: Responsible Plant Classifier
- Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.
- For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict, check and repair

REFERENCE · SECTION 20

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Use **residual** signs to say whether each plant-height estimate is too high or low.
- Which labelled step matches this detail? Mean **residual** is 1/3 cm; MAE is 5/3 cm, so small signed mean does not imply small errors.
- 1. Read directions
- 2. Avoid cancellation
- 3. Subtract matched pairs
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Avoid cancellation Mean **residual** is 1/3 cm; MAE is 5/3 cm, so small signed mean does not imply small errors. Read the picture in order.

Mean **residual** is 1/3 cm; MAE is 5/3 cm, so small signed mean does not imply small errors. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: Raise all **prediction**s by1 cm. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. **Residual**s become [-3, 0, 1]; mean=-2/3 cm and MAE=4/3 cm.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: **Residual**s become [-3, 0, 1]; mean=-2/3 cm and MAE=4/3 cm. Each **residual** decreases by1 under the observed-minus-predicted convention.

Repair and keep the record

Review Error Analysis (ai-4/18). Goal: Categorize development errors using **evidence**. Short practice: What **evidence** would support the claim that a change helps fresh dim-light images?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Accuracy and Imbalance (ai-4/16). Goal: Explain why overall accuracy can hide failure on a less common class. Short practice: Why can the 90% baseline still be unsuitable for a detection task?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Train a Classifier (ai-4/13). Goal: Connect **training**-only scaling and neighbour **prediction** in one pipeline. Short practice: Which pipeline record would help another team reproduce the result?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
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▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

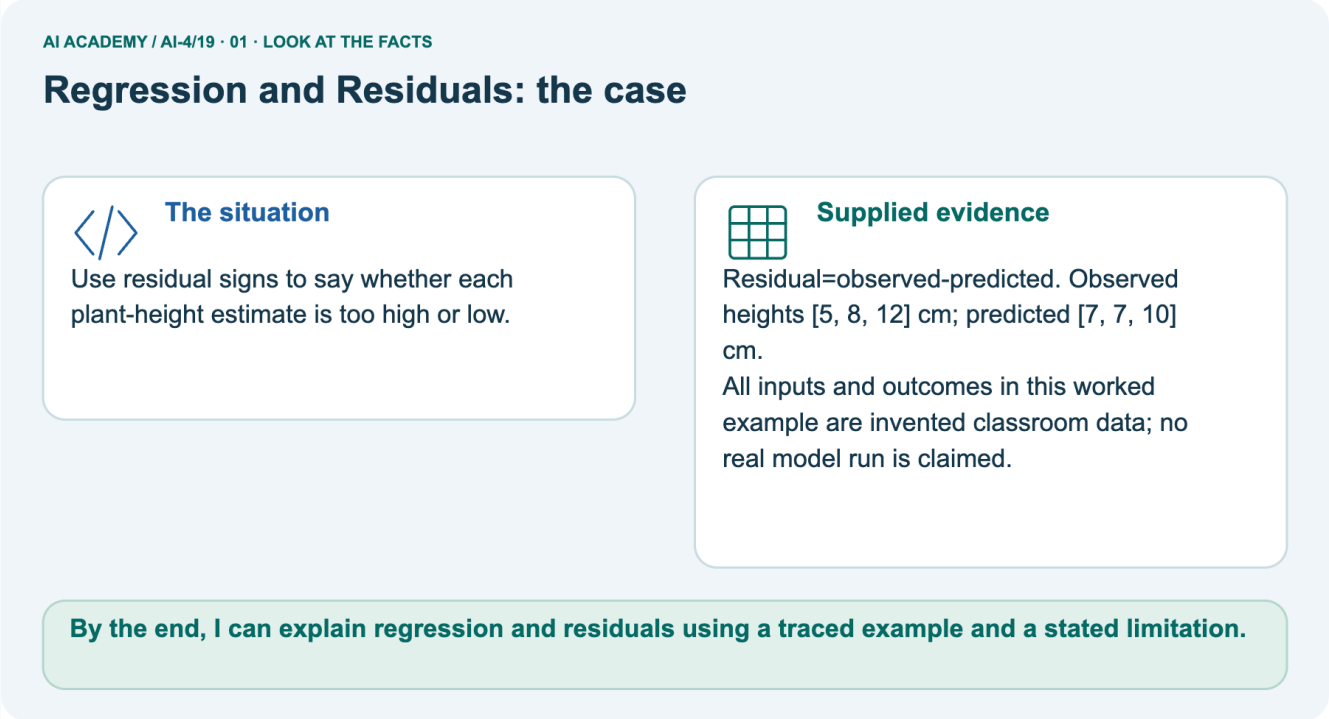
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the relevant picture instead of repeating every task.

01. Regression and Residuals: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

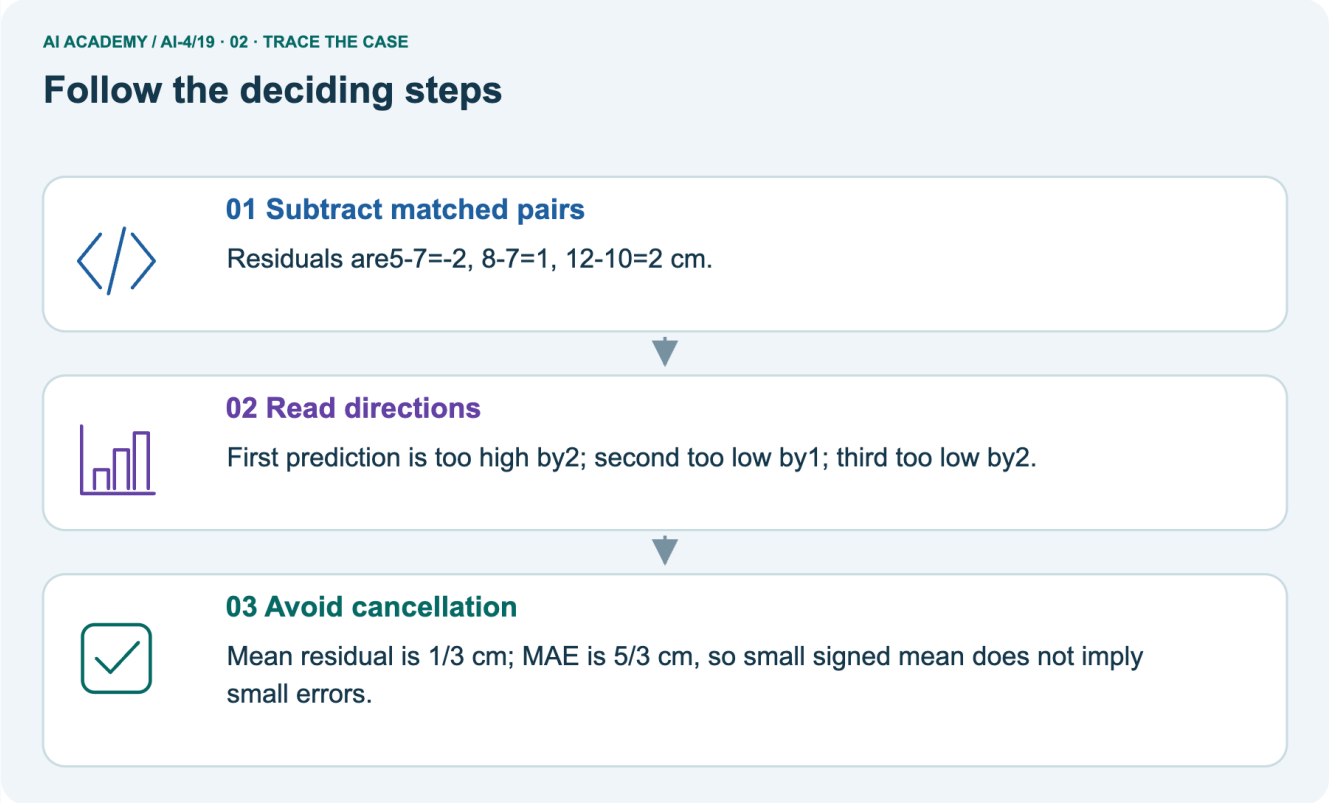
Use **residual** signs to say whether each plant-height estimate is too high or low. Supplied facts: **Residual**=observed-predicted. Observed heights [5, 8, 12] cm; predicted [7, 7, 10] cm.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



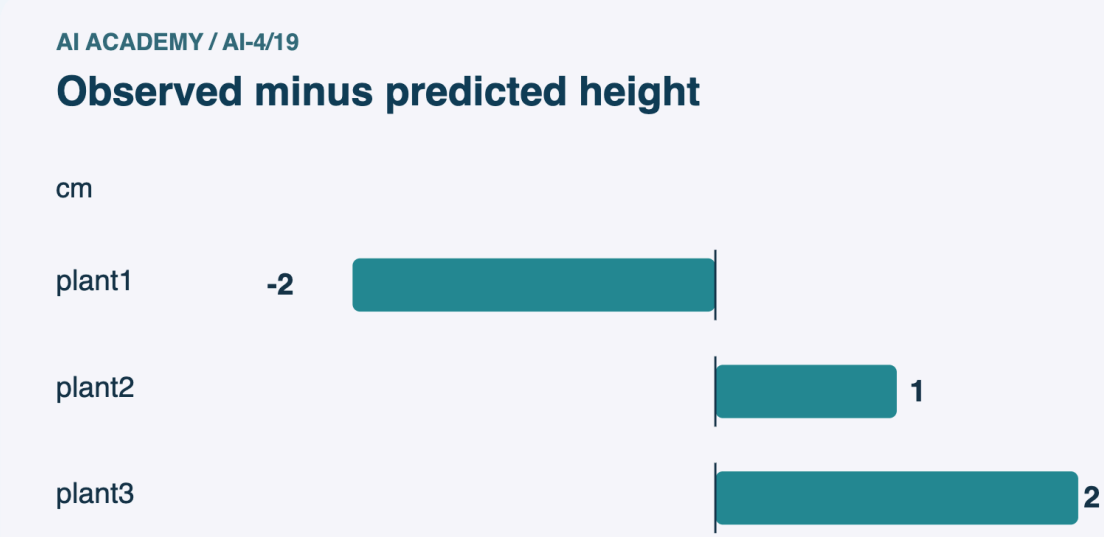
Follow the labelled picture with the matching explanation. Open the vector diagram

Subtract matched pairs: **Residual**s are5-7=-2, 8-7=1, 12-10=2 cm. → Read directions: First **prediction** is too high by2; second too low by1; third too low by2. → Avoid cancellation: Mean **residual** is 1/3 cm; MAE is 5/3 cm, so small signed mean does not imply small errors.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Observed minus predicted height



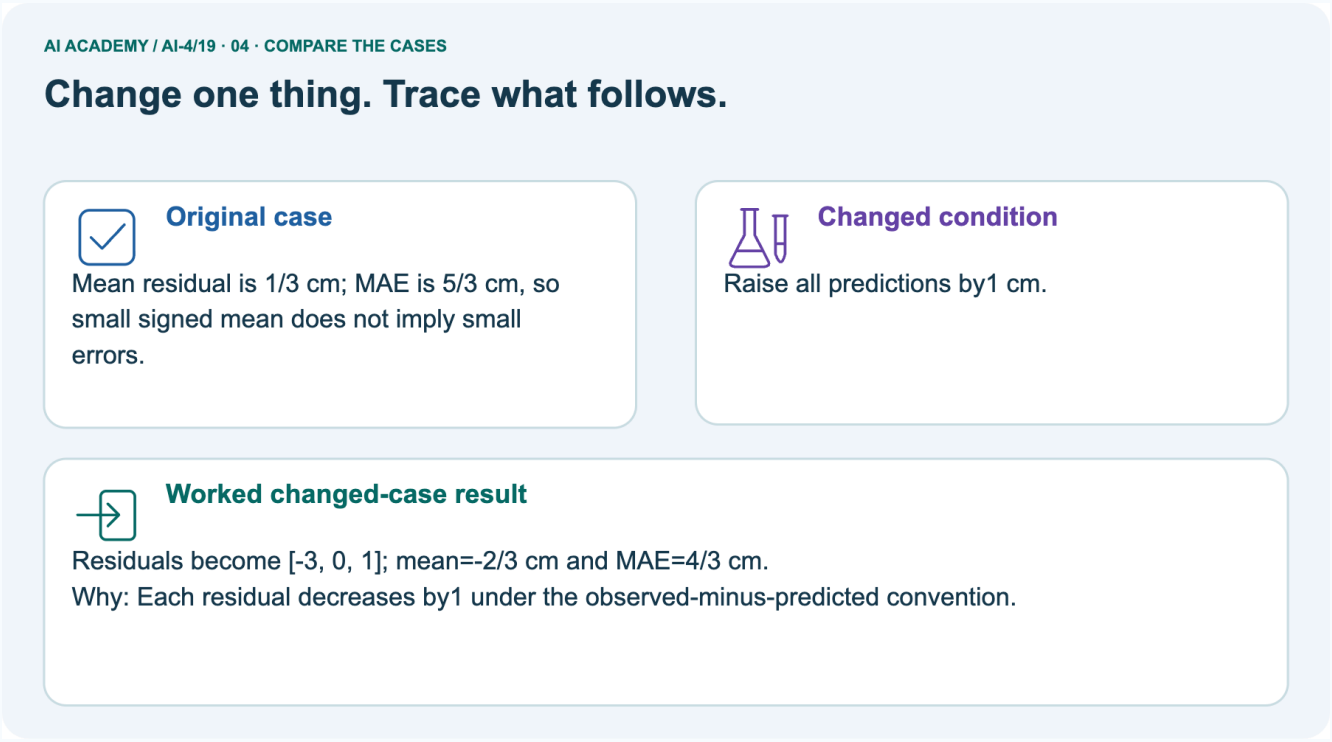
Follow the labelled picture with the matching explanation. Open the vector diagram

Observed minus predicted height. Units: cm. plant1: -2. plant2: 1. plant3: 2.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Mean **residual** is 1/3 cm; MAE is 5/3 cm, so small signed mean does not imply small errors. Changed condition: Raise all **prediction**s by1 cm.

Expected result: **Residual**s become [-3, 0, 1]; mean=-2/3 cm and MAE=4/3 cm. Reason: Each **residual** decreases by1 under the observed-minus-predicted convention.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit

AI ACADEMY / AI-4/19 · 05 · EXPLAIN WITH CARE

Connect the explanation to its limit

What the case explains

Read the picture in order. Mean residual is 1/3 cm; MAE is 5/3 cm, so small signed mean does not imply small errors. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What it does not establish

Always declare the residual sign convention; some software uses predicted minus observed.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Mean **residual** is 1/3 cm; MAE is 5/3 cm, so small signed mean does not imply small errors. The arrows show which recorded **input** or intermediate result each decision uses.

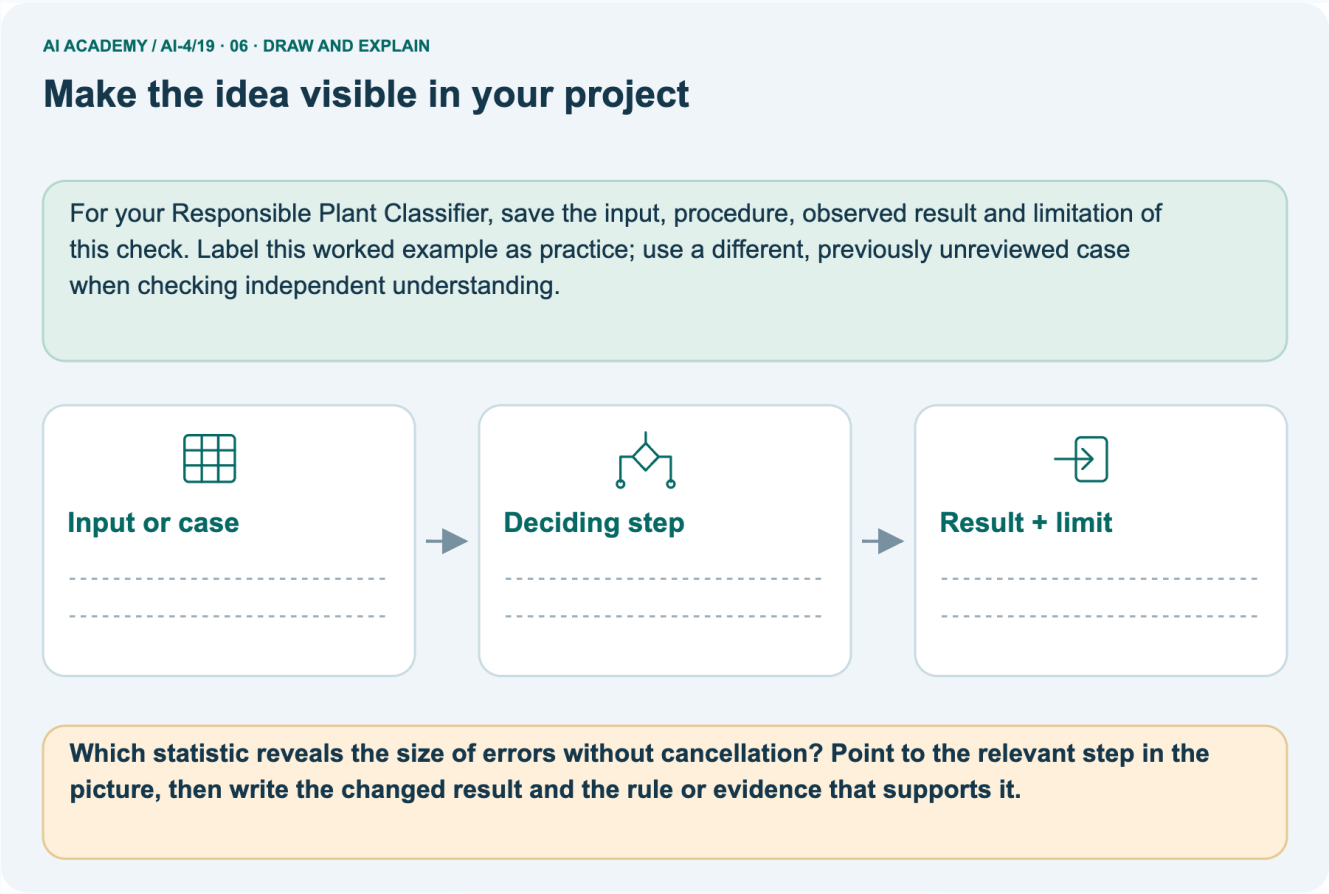
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: Always declare the **residual** sign convention; some software uses predicted minus observed.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which statistic reveals the size of errors without cancellation? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Regression and Residuals: the case

02. Follow the deciding steps

03. Observed minus predicted height

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.